

Computational Viability Certification under Incomplete Observation: A Continuous-to-Finite Bridge with Exact Certificates, a Solver-Assisted Certified Campaign, and a Finite-Side Reference Library

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Abstract

Nonviability certificates are useful computationally only if they are finite, attainable by standard solvers, and independently checkable. This paper develops such a pipeline for linear systems under exogenous partial observation of a hidden mode. An outer moment approximation of all measurable information-adapted controls, combined with an inner adversarial approximation (finitely stored admissible labels and scenarios), yields a finite linear program whose optimum is a certified lower bound on the continuous-time safety value and whose error inflation is a certified upper bound ($\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + \bar{e} + \delta_{\beta} + Lh_t$, $\bar{e}$ the program's own verified moment-error bound, $R_U V_U h_u$ in the uniform choice); every dual-feasible solution yields a finite lower-bound certificate covering every measurable policy, and a strictly positive certified margin is a finite obstruction certificate; along every admissible refinement sequence whose mesh and stored-scenario/enclosure widths vanish, nonviability is eventually detected, and whenever a positive certificate exists one can be chosen with at most $r + 1$ safety labels at rank r — the relevant dimension being information–time rank, not instantaneous input dimension, and a matching construction with scalar input produces obstructions whose minimal certificates require arbitrarily many witnesses, saturating the $r + 1$ bound. The certificates account for post-observation recourse: on the three-branch delayed-observation instance every pair of compatible states is viable (explicit pairwise policies are carried exactly) and a common control keeps every branch safe throughout the blind window, yet nonviability is certified, with the exact delay threshold $\tau^* = 7/50$; the certificate decodes into an exact shared-authority delay hierarchy — the triple is obstructed past $\tau = 7/50$, the pairs past ≈ 0.245 and ≈ 0.397 , singletons never — and the printed blind-window controls are exactly the optimal shared-braking programs. The certificate is prescriptive beyond the verdict: advancing observation buys margin at rate one, dilating post-revelation braking authority buys half a unit of delay tolerance per unit, and shared-window authority and the velocity budget buy exactly nothing; belief-cell rows extend the sandwich to partial and noisy observation with margins radius-bounded on both the witness and the certificate side. The finite program is solved by a standard solver (HiGHS) across a

five-point mesh refinement; optimality is certified in exact rational arithmetic by independent primal and dual witnesses, the solver agreeing with the exact values to at most 4.2×10^{-17} , and the refinement obeys the closed law $\rho(h) = 3/50 - Th/4$ (an unrestricted level, nonnegative on this family for $h \leq \frac{1}{5}$). A finite-side reference library, **viacert**, packages the exact finite-horizon recursion and the fibre-partition criteria behind a pluggable finite-system interface, with twelve audited identities in its self-test; the bridge's continuous programs live in the campaign scripts, not in the library. All exact verification paths for the reported rational benchmark run in integer and rational arithmetic; solver outputs are numerical confirmations, never proof objects.

Keywords: viability; incomplete observation; obstruction certificates; linear programming; delayed observation; exact arithmetic; certified computation

1. Introduction

Viability theory under full observation has a mature numerical apparatus: kernel approximations, set-valued meshes, and convergence results (Aubin, 1991; Cardaliaguet, Quincampoix, and Saint-Pierre, 1999). Under incomplete observation the sufficiency side is carried by output-feedback conditions and the estimation-tube program (Veliov, 1993), while the necessity side — certificates whose firing *proves* nonviability of an information state (Abaee, 2026b) — has lacked a computational bridge. The difficulty is the quantifier order: the adversary chooses the mode, the initial state inside its cell, and a measurable disturbance with knowledge of the announced policy, so naive discretization of controls proves at most the infeasibility of a held-control subclass, not of the policy class itself.

This paper builds the bridge and the tooling around it. The main result (Theorem 4) approximates the admissible controls from outside by their block moments — a necessary relaxation of continuous-time safety that restricts nothing — and the adversary from inside by explicitly stored scenarios, yielding a finite linear program whose optimal value sandwiches the exact minimax safety violation. Dual solutions of the program are finite, policy-independent obstruction certificates; mesh refinement eventually certifies — on a nonviable system suffi-

ciently fine meshes produce a positive certificate, though positivity is not itself monotone under refinement (Theorem 4(iii)); and their size is governed by the rank of the safety-row matrix in the information–time product, which no bound on the instantaneous input dimension controls (Proposition 5).

The pipeline is demonstrated end to end on a three-branch delayed-observation instance (the running instance of this paper; Section 6): delay threshold, Farkas certificate, and a five-point mesh campaign in which the HiGHS solver (Huangfu and Hall, 2018) agrees with the exact rational optimum at every mesh to floating-point tolerance, with independent primal–dual witnesses closing the duality gap exactly. The finite-side machinery — the exact backward recursion and the fibre-partition criteria — is packaged as the reference library `viacert` with a pluggable finite-system interface and a self-test of twelve audited identities; the library supplies the enumerate-and-check layer of the certification workflow, and it is deliberately not a reimplementaion of the bridge’s continuous programs (Section 11).

The contributions are: the continuous-to-finite sandwich theorem with sound, refinement-complete, sparsity-bounded certificates (Section 5); the recourse-aware obstruction reading and the exact three-branch instance with its certified campaign (Section 6); the information–time rank separation (Section 8); the refinement-erosion identity, the oracle-recourse extension, and the belief-cell certificates for partial and noisy observation (Section 9); the certificate’s prescriptive decode into optimal shared braking authorities, the induced pair-delay hierarchy, and the exact redesign sensitivities (Section 10); the complexity and residual-bounded verification framework (Section 7); and the reference library (Section 11).

2. Model, information, and quantifiers

Dynamics. Let the hidden mode be $\theta \in \Theta := \{1, \dots, M\}$. Conditional on θ , the continuous state satisfies

$$\dot{x}(t) = A_\theta(t)x(t) + g_\theta(t) + B_\theta(t)u(t) + E_\theta(t)d(t), \quad t \in [0, T],$$

under the standing assumptions: $A_\theta, g_\theta, B_\theta, E_\theta$ bounded and piecewise continuous, with bounded-variation bounds on B_θ, E_θ — the operative assumption below is the induced bound $\text{TV}(k_a) \leq V_U$ on the adjoint kernels, assumed directly in Theorem 4 rather than derived from the primitive coefficients here; the initial set in mode θ a nonempty compact polytope $P_\theta \subset \mathbb{R}^n$; the input and disturbance sets nonempty compact polytopes $U = \{v : Fv \leq f\} \subset \mathbb{R}^m$ and $D_\theta \subset \mathbb{R}^p$; safety the closed polyhedron $\mathcal{V} = \{x : c_i^\top x \leq b_i, i = 1, \dots, s\}$; and the initial information state $B_0 = \bigcup_\theta \{\theta\} \times P_\theta$ initially safe, meaning $P_\theta \subseteq \mathcal{V}$ for every θ (if some P_θ leaves $\mathcal{V}$ at time zero, a time-zero certificate suffices and the analysis is vacuous). The dynamics admit unique Carathéodory solutions for every admissible measurable signal pair.

Information. At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R = T$ the controller receives exogenous information about θ : refining history partitions $\mathcal{P}_0 \preceq \mathcal{P}_1 \preceq \dots \preceq \mathcal{P}_{R-1}$ of Θ — the order is by refinement: for $r' > r$ every cell of $\mathcal{P}_{r'}$ lies inside a cell of $\mathcal{P}_r$ (the last partition is $\mathcal{P}_{R-1}$ because $\tau_R = T$ ends the last interval) — the observation history during $I_r = [\tau_r, \tau_{r+1})$ identifying the cell $C_r(\theta) \in \mathcal{P}_r$.

These are history partitions, not necessarily partitions generated by the latest observation alone. Observations do not depend on x, u, d ; active sensing and endogenous observation branching are excluded from the exact theorem (Section 9 gives a sound extension).

Policies and quantifiers. Policies are deterministic, measurable, nonanticipative, and adapted to the observation history; every such policy is represented exactly by signals $u_{r,C} \in L^\infty(I_r; U)$, $C \in \mathcal{P}_r$ — the realized control in mode θ being $u_\theta^\pi(t) = u_{r,C_r(\theta)}(t)$ on I_r . The robust quantifier order is

$$\exists \pi \in \Pi_{\mathcal{I}} \forall \theta \forall x_0 \in P_\theta \forall d \in L^\infty([0, T]; D_\theta) \forall t \in [0, T],$$

the adversary selecting mode, initial state, and measurable disturbance with knowledge of the announced policy; randomized policies are not included. Throughout, a belief is a nonempty set $B \subseteq \Theta \times \mathbb{R}^n$ of mode-state possibilities, not a probability distribution. The information-constrained viability kernel is $\mathcal{K}_{\mathcal{I}}^T = \{B : \exists \pi \forall \theta, x_0, d, t \ x_{\theta, x_0, d}^\pi(t) \in \mathcal{V}\}$, a set of beliefs; the audited initial state B_0 is viable exactly when $B_0 \in \mathcal{K}_{\mathcal{I}}^T$. Writing $\text{Lift}(K_{\text{full}}^T) = \{B \subseteq \Theta \times \mathbb{R}^n : \emptyset \neq B \subseteq K_{\text{full}}^T\}$ for the belief lifting of the full-information kernel (an ambient $\Theta \times \mathbb{R}^n$ -valued family), $\mathcal{K}_{\mathcal{I}}^T \subseteq \text{Lift}(K_{\text{full}}^T)$, and the inclusion is strict in general: the instance of Section 6 has every two-state subbelief viable — while the three-state belief is not.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
$\Gamma_{\mathcal{I}}(\lambda, a)$: the finite obstruction certificate margin (1)	the recourse certificate Γ_h (obstruction calculus; minimax duals); the alpha-vector witness sets Γ_k^{Π} (sufficiency)
λ_ℓ : the certificate weights	the Farkas multiplier and observer decay rate (obstruction calculus); the Farkas and dual weights (worked systems)
μ : the adversarial probability measure	Farkas multipliers (obstruction calculus); the adversarial measures of the minimax companion
K_{full}^T : the full-information viability kernel	the safe set and kernel K (obstruction calculus); the window depth and contact set $\mathcal{K}(B)$ (minimax duals); stock carrying capacities (forecast ladder)
ρ : the mesh relaxation parameter	the distributionally robust mixture parameter ρ (sufficiency; minimax duals); the net biomass multiplier (applied regime viability)
β_a : the stored per-label bound	the margin β (minimax duals)
τ : the observation time	the worst-case exit time τ^* (worked systems); the delay times τ (obstruction calculus)
σ : the kernel time variable	the floored exit count σ^* (worked systems); a noise scale (forecast ladder)

3. Adjoint safety rows and the safety value

Let $X_\theta(t, s)$ denote the state-transition matrix. For a label $a = (\theta, i, t) \in \mathcal{S} := \Theta \times \{1, \dots, s\} \times [0, T]$ define the input kernel and margin

$$k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i,$$

$$\begin{aligned} \beta_a &= b_i - h_{P_\theta}(X_\theta(t, 0)^\top c_i) - \int_0^t c_i^\top X_\theta(t, \sigma) g_\theta(\sigma) d\sigma \\ &\quad - \int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma, \end{aligned}$$

where $h_Q(v) = \sup_{q \in Q} v^\top q$. For a policy π set $\mathcal{F}_a(\pi) = \int_0^T k_a(\sigma)^\top u_\theta^\pi(\sigma) d\sigma - \beta_a$.

Proposition 1 (robust rows). *π is robustly safe on $[0, T]$ if and only if $\mathcal{F}_a(\pi) \leq 0$ for every $a \in \mathcal{S}$.*

Proof. The identity is the linear solution formula: the initial-state and disturbance maximizations separate from the control term because the control signal in a mode does not depend on x_0 or d , so the rows are exact. For polytopic D_θ , a maximizing disturbance can be chosen measurably by selecting a maximizing vertex at each time, ties resolved by a fixed ordering. $\square$

Define the minimax safety violation $J_{\mathcal{I}}(T) = \min_{\pi \in \Pi_{\mathcal{I}}} \max_{a \in \mathcal{S}} \mathcal{F}_a(\pi)$.

Proposition 2 (value and nonviability). *The minimum exists: the policy-signal space is a finite product of weak-* compact sets $L^\infty(I_r; U)$ (each U a compact convex polytope, so the admissible signal sets are weak-* closed and bounded) and each $\mathcal{F}_a$ is weak-* continuous; the maximum over labels is attained on the compact label set $\mathcal{S}$ because $a \mapsto \mathcal{F}_a(\pi)$ is continuous in t under assumption (6) below. Consequently $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$ if and only if $J_{\mathcal{I}}(T) > 0$.*

Proof. Existence: the policy-signal space is a finite product of weak-* compact sets $L^\infty(I_r; U)$ (each U a compact convex polytope, so the admissible signal sets are weak-* closed and bounded) and each $\mathcal{F}_a$ is weak-* continuous; the maximum over labels is attained on the compact label set $\mathcal{S}$ because $a \mapsto \mathcal{F}_a(\pi)$ is continuous in t under assumption (6) below. The equivalence is then the definition of the kernel: a violation-free policy exists exactly when the minimax value vanishes. The strict positive margin is a consequence of compactness and closed safety constraints, not an extra nonviability assumption. $\square$

4. The finite obstruction certificate

Define $\phi_U(g) := \min_{v \in U} g^\top v = -h_U(-g)$. Choose finitely many labels $a_\ell = (\theta_\ell, i_\ell, t_\ell)$, $\ell = 1, \dots, q$, and weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$. For each active information cell $C \in \mathcal{P}_r$, pool the adjoint kernels $g_{r,C}(\sigma) = \sum_{\ell: \theta_\ell \in C} \lambda_\ell k_{a_\ell}(\sigma)$. The certificate margin is

$$\Gamma_{\mathcal{I}}(\lambda, a) = \sum_{r=0}^{R-1} \int_{I_r} \sum_{C \in \mathcal{P}_r} \phi_U(g_{r,C}(\sigma)) d\sigma - \sum_{\ell=1}^q \lambda_\ell \beta_{a_\ell}. \quad (1)$$

Remark 3 (semantics). If $\Gamma_{\mathcal{I}}(\lambda, a) > 0$, then no admissible information-adapted policy is safe: for every policy,

$\sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a)$, so at least one stored label satisfies $\mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a) > 0$ — an admissible realization violates facet i_ℓ at time t_ℓ . The violating label may depend on the policy; the finite list and its weights do not. The certificate is not an adversarial tree: it is a finite, policy-independent set of adversarial tests combined by an adjoint identity. The pooled bound is tight as a minimum over policies: $\min_\pi \sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) = \Gamma_{\mathcal{I}}(\lambda, a)$, the pointwise minimizer being measurable by fixed-order vertex selection in the polytope U .

5. The continuous-to-finite theorem

The exact dual expression (1) is a genuine infinite-dimensional minimax statement, and we prove it in full rather than invoke it: the proof of (iv) below (complete version in the Supplementary Material) runs Sion's minimax theorem over the mixed extension to probability measures on the label set and transfers the result to finitely supported measures by weak-* density. The consequential result is a sound finite construction covering unrestricted measurable controls; classical infinite-dimensional duality (Anderson and Nash, 1987; Hettich and Kortanek, 1993) is the surrounding theory, not a load-bearing citation.

Outer control moments. Choose a control-time mesh $\mathcal{H} = \{J_1, \dots, J_H\}$ containing every sensing time. Introduce one vector $v_{j,C} \in U$ for every cell $C \in \mathcal{P}_{r(j)}$ active during J_j , where $r(j)$ is the unique observation interval containing J_j (the mesh contains every sensing time, so each block lies in one I_r), and let $Q = \sum_j |\mathcal{P}_{r(j)}|$ be the number of information-time blocks. For a label a define the finite row $A_{a,j,C} = \mathbf{1}_{\{\theta \in C\}} \int_{J_j} k_a(\sigma) d\sigma$, the block average $\bar{k}_{a,\mathcal{H}}(\sigma) = \frac{1}{|J_j|} \int_{J_j} k_a(s) ds$ on J_j , and the moment error

$$e_{a,\mathcal{H}} = \int_0^T h_U(\bar{k}_{a,\mathcal{H}}(\sigma) - k_a(\sigma)) d\sigma, \quad (2)$$

for which any verified upper bound $\bar{e}_a \geq e_{a,\mathcal{H}}$ may be used. For every measurable policy, its block averages belong to U , and

$$\sum_{j,C} A_{a,j,C}^\top v_{j,C} - \int_0^T k_a(\sigma)^\top u_{\theta(a)}^\pi(\sigma) d\sigma \leq \bar{e}_a. \quad (3)$$

Inequality (3) is the crucial orientation: finite feasibility is a necessary condition for continuous-time safety. The moments do not restrict the real controller to held inputs. For the sampled-data subclass whose policies are held constant on the blocks, the averaging term vanishes identically ($\int_{J_j} (k_a - \bar{k}_{a,\mathcal{H}}) d\sigma = 0$ on every block), so such a policy is continuously safe exactly when its block moments satisfy the unrelaxed rows: an exact reduction precisely when the row bounds are exact, $\bar{e}_a = 0$; with the error bounds (2) the program remains a relaxation even for held policies (in the instance the held witness policy attains the continuous value $\frac{3}{50}$ on every aligned mesh, while (5) returns $\frac{3}{50} - Th/4$).

Inner adversarial approximation. One may use a right-hand side $\beta_a^+ \geq \beta_a$ obtained from an explicit admissible initial state and disturbance signal rather than exact worst-case optimization. If $0 \leq \beta_a^+ - \beta_a \leq \delta_\beta$ uniformly over the labels used, the approximation is safe for nonviability: it weakens the adversary and enlarges the feasible controller set. For example, with

$w_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$, on a disturbance mesh choose $d_{a,J} \in \arg \max_{d \in D_\theta} (\int_J w_a)^\top d$, yielding an explicit piecewise-constant admissible disturbance; if D_θ has radius R_D about a fixed center and $\text{TV}(w_a) \leq V_D$, then

$$0 \leq \beta_a^+ - \beta_a \leq R_D V_D h_d, \quad (4)$$

the general budget being $\delta_\beta = \delta_{\text{init}} + \delta_{\text{dist}} + \delta_{\text{num}}$ for separately accounted initial-state and numerical error.

The finite program. Let $G \subset [0, T]$ be a finite time grid with $0, T \in G$ and every observation time in G ; the label set is the product $\mathcal{S}_G = \Theta \times \{1, \dots, s\} \times G$, with covering radius $h_t = \sup_{t \in [0, T]} \min_{\hat{t} \in G} |t - \hat{t}|$. Solve

$$\rho_{\mathcal{H}, G} = \min_{v, \rho} \rho \quad \text{s.t.} \quad A_a v - \beta_a^+ - \bar{e}_a \leq \rho \\ (a \in \mathcal{S}_G), v_{j,C} \in U. \quad (5)$$

The level ρ is an unrestricted real variable; negative values record strict safety slack.

Theorem 4 (continuous-to-finite). *Assume:*

- (H1) *uniformly over admissible policies, modes, and facets, $|\mathcal{F}_{\theta, i, t}(\pi) - \mathcal{F}_{\theta, i, t'}(\pi)| \leq L|t - t'|$ (6);*
- (H2) *the input kernels satisfy $\text{TV}(k_a) \leq V_U$ (7);*
- (H3) *the label maps $a \mapsto k_a$ and $a \mapsto \beta_a$ are continuous on the compact label set (k_a in L^1 norm, β_a absolutely; the L^1 form is the operative one — sup-norm continuity is neither needed nor true here, the step kernels below jumping at $\sigma = t$ for every t — while L^1 continuity is implied by bounded coefficients and enjoyed by every rational instance below);*
- (H4) *U has radius R_U about some fixed center;*
- (H5) *the time grid has covering radius at most h_t and the control mesh maximum length is h_u ;*
- (H6) *the adversarial approximation satisfies the bound above.*

Then:

(i) *Certified value sandwich.*

$$\rho_{\mathcal{H}, G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H}, G} + \max_{a \in \mathcal{S}_G} \bar{e}_a + \delta_\beta + Lh_t. \quad (8)$$

With (4) — the budget $\delta_\beta = \delta_{\text{init}} + \delta_{\text{dist}} + \delta_{\text{num}}$ — $J_{\mathcal{I}}(T) - \rho_{\mathcal{H}, G} \leq \max_a \bar{e}_a + R_D V_D h_d + \delta_{\text{init}} + \delta_{\text{num}} + Lh_t$. Verified numerical row widths are added to the right side. The upper bound is realized by held controls, for which the averaging error vanishes identically, so the bound is charged by the error bounds the program actually used; the uniform choice $\bar{e}_a \equiv R_U V_U h_u$ — valid by (2) whenever $\text{TV}(k_a) \leq V_U$, since $\int_J |k_a - \bar{k}_a| \leq h_u \text{TV}(k_a|_J)$ and $h_U(g) \leq R_U \|g\|$ — recovers the announced $R_U V_U h_u$ form as a special case. The $\max \bar{e}_a$ form is the conservative one: a loose but verified choice inflates ρ 's relaxation and must inflate the charged bound with it (a counterexample is carried in Section 6).

(ii) *Sound finite certificates.* If $\rho_{\mathcal{H}, G} > 0$, the program's dual supplies a finite certificate of nonviability of the original continuous-time system, covering every measurable information-adapted policy.

(iii) *Completeness under refinement.* If $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$, then for every admissible refinement sequence along which

h_u, h_d, h_t converge to zero and the stored-scenario widths $\delta_{\text{init}}, \delta_{\text{num}}$ do as well — near-maximizing stored initial states (Section 7) must be refined with the mesh for completeness to hold — the meshes eventually produce a positive finite certificate. Nonviability is thus semidecidable in this class for the input model in which the coefficients are exact rational or interval-enclosed piecewise-polynomial data with certified integration, total-variation, and uniform time-Lipschitz bounds: the semidecision procedure enumerates admissible meshes, solves each enclosed rational program, and halts on the first positive certificate — for instance, the dyadic schedule $h_u = h_d = h_t = T/2^k$ halts with certificate margin at least $\gamma/2$ once $2^{-k}(R_U V_U + R_D V_D + L)T < \gamma/2$ (with exactly stored scenarios, $\delta_{\text{init}} = \delta_{\text{num}} = 0$), where $\gamma = J_{\mathcal{I}}(T) > 0$ is the continuous value.

(iv) *Exact atomic dual representation.*

$$J_{\mathcal{I}}(T) = \sup_{q < \infty, a_1, \dots, a_q, \lambda \in \Delta_q} \Gamma_{\mathcal{I}}(\lambda, a). \quad (9)$$

In particular, nonviability is equivalent to the existence of a positive finite atomic certificate.

(v) *Sparse witnesses.* Let $r = \text{rank } A$ for the finite safety-row coefficient matrix of (5) — the A_a rows excluding the ρ -column and the actuator inequalities; the bound is linear-programming (LP) basic-solution theory (Chvátal, 1983; supplementary Step 6). Whenever $\rho_{\mathcal{H}, G} > 0$, a positive certificate exists using at most $r + 1 \leq mQ + 1$ safety labels, and it can retain any prescribed margin strictly below $\rho_{\mathcal{H}, G}$. If the lifted continuous-time kernels span a finite-dimensional space of dimension $r_{\mathcal{I}}$, a continuous-time obstruction admits one with at most $r_{\mathcal{I}} + 1$ labels. The relevant dimension is information-time rank, not m .

Proof (the complete steps are Steps 1–6 of the supplementary material, verified by the verification script). The lower bound holds because every admissible measurable policy's control bounds are outer-approximated by the block moments, so the moment program relaxes the controller's feasible set and its value cannot exceed the continuous safety value; the upper bound is realized by taking, on each block, the piecewise-constant control attained by the finite optimizer's block variables, whose exit from the continuous value is charged exactly by the program's own moment bounds $\bar{e}_a$ ($R_U V_U h_u$ under the uniform choice), the stored-scenario error δ_β , and the time-grid interpolation error Lh_t . The asymmetry at its core is worth stating plainly: averaging arbitrary controls proves the lower bound; implementing piecewise-constant controls proves the upper bound. Confusing these two directions is what makes naïve infeasibility-by-discretization arguments unsound.

Proof of (iv) (exact atomic dual representation). The policy space $\Pi_{\mathcal{I}} = \prod_{r,C} \{u \in L^\infty(I_r; U)\}$ is weak-* compact and convex (Banach-Alaoglu: each factor is weak-* closed and bounded, U being a compact convex polytope), and the label set $\mathcal{S}$ is compact. Under the label-continuity hypothesis, $(a, \pi) \mapsto \mathcal{F}_a(\pi)$ is jointly continuous (π -continuity is the L^1 -pairing; a -continuity is uniform in π by the L^1 label continuity and boundedness of the U -values: $|\mathcal{F}_a(\pi) - \mathcal{F}_{a'}(\pi)| \leq M_U \|k_a - k_{a'}\|_{L^1} + |\beta_a - \beta_{a'}|$ with $M_U := \sup_{v \in U} \|v\|$ (Hölder; the controls are M_U -bounded), the equicontinuity Arzelà-Ascoli consumes). Let $\mathcal{P}(\mathcal{S})$ carry the weak-* topology: by the Riesz representation it is the compact convex metrizable set of

Borel probability measures. Since the integral of a continuous function over a probability measure lies within its min-max range and the Dirac measures attain both ends, $\max_a \mathcal{F}_a(\pi) = \max_\mu \int \mathcal{F}_a(\pi) d\mu$, and the bilinear $\int \mathcal{F}_a(\pi) d\mu$ — affine, hence simultaneously quasi-convex and quasi-concave, and continuous in each variable — meets the hypotheses of Sion's minimax theorem, giving

$$J_{\mathcal{I}}(T) = \max_{\mu \in \mathcal{P}(\mathcal{S})} \min_{\pi} \int \mathcal{F}_a(\pi) d\mu,$$

the outer maximum attained. The value function $\mu \mapsto \min_{\pi} \int \mathcal{F}_a(\pi) d\mu$ is weak-* continuous: the family $\{a \mapsto \mathcal{F}_a(\pi)\}_{\pi}$ is uniformly bounded and uniformly equicontinuous, so by Arzelà–Ascoli weak-* convergence of measures makes the pairings converge uniformly over π . Finitely supported measures are weak-* dense in $\mathcal{P}(\mathcal{S})$, so this continuous value function has the same supremum over finitely supported measures as over all of $\mathcal{P}(\mathcal{S})$ — namely $J_{\mathcal{I}}(T)$. For $\mu = \sum_{\ell} \lambda_{\ell} \delta_{a_{\ell}}$ the inner minimum decomposes over information cells, and the cellwise infimum of an L^1 -pairing over U -valued signals equals the pointwise ϕ_U integral by measurable vertex selection (fixed vertex order in the polytope), which is exactly the representation $\Gamma_{\mathcal{I}}(\lambda, a)$ of (1). Nonviability $J_{\mathcal{I}}(T) > 0$ therefore yields a finitely supported μ with $\Gamma > J_{\mathcal{I}}(T)/2 > 0$; conversely $\Gamma > 0$ lower-bounds $J_{\mathcal{I}}(T)$ because $\max_a \geq$ the λ -weighted average. The complete seven-step verification of the remaining hypotheses is Step 5b of the Supplementary Material; the label-continuity input is the L^1 estimate above, verified in Section 6 for the audited kernels.

Corollary[complete dual calculus]. Under the hypotheses of Theorem 4: (a) every dual-feasible pair (λ, a) satisfies $\Gamma_{\mathcal{I}}(\lambda, a) \leq J_{\mathcal{I}}(T) \leq \max_a \mathcal{F}_a(\pi)$ for every policy — the lower bound is Remark 3, the upper is the weighted-average bound of the proof; (b) $J_{\mathcal{I}}(T) = \sup_{q, a, \lambda \in \Delta_d} \Gamma_{\mathcal{I}}(\lambda, a)$: the supremum over all measures is attained (Sion), and finitely supported measures approach its value arbitrarily finely — attainment by an atomic maximizer is an instance-level fact (in the instance of Section 6 three atoms attain it), not a general claim; (c) $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$ if and only if some dual-feasible pair has $\Gamma_{\mathcal{I}}(\lambda, a) > 0$; every such pair obstructs every measurable information-adapted policy, and every obstruction arises this way. The duals therefore form the complete certificate system for the class, positivity being the only decision.

6. A three-branch delayed-observation instance carried exactly

Plant and exact delay threshold. Consider the planar double integrator $\dot{p} = v$, $\dot{v} = u$, with unit vectors $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$, the hexagonal input set $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$, equivalently $\{v : |v_2| \leq \frac{4}{5}, |2u_1 + u_2| \leq 2, |2u_1 - u_2| \leq 2\}$, safety $n_i^\top p \leq 2$ ($i = 1, 2, 3$) and $|v_1|, |v_2| \leq \frac{6}{5}$ — seven facets in all, $s = 7$ (three position, four velocity) — three possible initial states $p_j(0) = \frac{34}{25}n_j$, $v_j(0) = n_j$, no disturbance, and the mode unobserved until time τ , when it is revealed exactly.

If j is known, $u = -n_j$ for one second and then $u = 0$ keeps the maximum critical position at $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} = 1.86 < 2$: all three physical states are indefinitely viable under full information (throughout the braking leg

$\|v(t)\|_{\infty} \leq 1 \leq \frac{6}{5}$; the noncritical facets keep $n_i^\top p < 0$ along the n_j -ray; $-n_j \in U$). Take three witness labels, each using its own position facet at $t_* = \tau + 1$, with kernels $k_j(s) = \mathbf{1}_{\{s \leq t_*\}}(t_* - s)n_j$ and $\beta_j = \frac{16}{25} - t_*$, weighted by $\lambda_1 = \frac{3}{8}$, $\lambda_2 = \lambda_3 = \frac{5}{16}$, which satisfy $\sum_j \lambda_j = 1$ and $\sum_j \lambda_j n_j = 0$. Before observation the pooled kernel is identically zero; after observation each mode has its own control and $\phi_U(\lambda_j(t_* - s)n_j) = -\lambda_j(t_* - s)$. Hence

$$\Gamma(\tau) = - \int_{\tau}^{\tau+1} (\tau+1-s) ds - \left(\frac{16}{25} - \tau - 1\right) = \tau - \frac{7}{50}, \quad (10)$$

so $\tau > \frac{7}{50}$ implies $B_0 \notin \mathcal{K}_{\mathcal{I}}$. Conversely, $u = 0$ before observation and $u = -n_j$ for one second afterward keeps the maximum critical position at $\frac{34}{25} + \tau + \frac{1}{2}$, safe exactly when $\tau \leq 0.14$. Thus $\tau_{\max} = \frac{7}{50} = 0.14$ is exact (equality viable: the safety set is closed), and for horizons $T \geq \tau + 1$ the upper-bound policy — full braking from τ on, stopping at $\tau + 1$ from $p = (\frac{34}{25} + \tau)n_j$ — shows every row value at most $\tau - \frac{7}{50}$, so $J_{\mathcal{I}}(T) = \tau - \frac{7}{50}$.

Pairs are viable; the blind-window exit test misses. At $\tau = \frac{1}{5}$, the common control $u = 0$ keeps all three branches safe throughout the blind window: $n_j^\top p_j(0.2) = \frac{39}{25} = 1.56 < 2$. The pair claims are carried exactly. For $\{1, 2\}$ (and $\{1, 3\}$ by the reflection $v_2 \mapsto -v_2$): brake *through* the blind window with the common control $u = -(n_1 + n_2) = (-\frac{2}{5}, -\frac{4}{5}) \in U$ (both branches decelerate along their normals at $\frac{2}{5}$), then brake each branch along its own normal, $u = -n_j$ from τ to n_j -rest — the critical position peaks at $\frac{12345}{6250} = 1.9752 < 2$ on both branches, slack $\frac{31}{1250}$. For $\{2, 3\}$: hold $u = n_1$ through the blind window (both branches decelerate at $\frac{3}{5}$), then brake each branch along its own normal, $u = -n_j$ from τ — the critical position peaks at $\frac{2419}{1250} = 1.9352$, slack $\frac{81}{1250}$. (Two tempting variants fail: holding $u = 0$ through the blind window in $\{1, 2\}$ peaks at $\frac{2719}{1250} = 2.175 > 2$ with the printed mid-brake, and at $\frac{103}{50}$ even with full braking afterward; inserting a $-\frac{3}{5}n_j$ leg after τ in $\{2, 3\}$ peaks at $\frac{2501}{1250} = 2.0008 > 2$ at $t = \frac{29}{25}$.) All seven facets hold at every time on every in-belief branch of every pair policy on $[0, \frac{6}{5}]$: every velocity value along $\{1, 2\}$ satisfies $\|v\|_{\infty} \leq 1$ and along $\{2, 3\}$ $\|v\|_{\infty} \leq \frac{4}{5}$, both strict against $\frac{6}{5}$, and the worst slacks are the peak rows $\frac{31}{1250}$ and $\frac{81}{1250}$; braking to n_j -rest leaves a harmless transverse velocity of $\frac{4}{25}$ (its drift only decreases the noncritical coordinates). The blind control $u = n_1$ does drive the off-belief branch 1 to $|v_1| = \frac{6}{5}$ exactly at τ — irrelevant to the pair claim (branch 1 is not in the belief), and indeed that control could not keep branch 1 safe: its peak would be $\frac{23}{10} = 2.3 > 2$. No exit occurs before information arrives; nevertheless $\Gamma(0.2) = 0.06 > 0$ — the best post-observation recourse cannot recover all branches. The obstruction is not that information arrives after unavoidable physical exit; it is that information arrives after some indistinguishable branch has necessarily accumulated more braking liability than the remaining control authority can discharge.

The finite program and its exact witnesses. Set $\tau = \frac{1}{5}$, $T = \frac{6}{5}$, $h = \frac{1}{10}$: two common pre-observation control blocks and ten post-observation blocks for each of three modes, so $Q = 32$, $mQ = 64$. For the endpoint position rows the exact moment error is $e = 12 \cdot h^2/4 = \frac{3}{100}$ (the kernel is affine on each of the $T/h = 12$ mesh intervals and $h_U(\alpha n_j) = |\alpha|$); the endpoint right-hand side is $\beta = \frac{16}{25} - \frac{6}{5} = -\frac{14}{25}$, relaxed to $\beta + e = -\frac{53}{100}$. The ex-

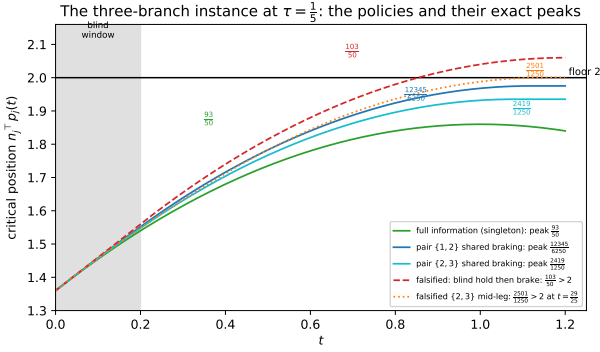


Figure 1: The audited policies on the critical coordinate $n_j^\top p_j(t)$ at $\tau = \frac{1}{5}$, every peak the exact rational of the text: the full-information singleton peaks at $\frac{93}{50}$; the pair policies peak below the floor ($\frac{12345}{6250}$, $\frac{2419}{1250}$, slacks $\frac{31}{1250}$ and $\frac{81}{1250}$); the two falsified variants cross it ($\frac{103}{50}$, $\frac{2501}{1250}$ at $t = \frac{29}{25}$). Generated by the assertion script `paper2_comp_certification_figures_v1.py`.

act continuous certificate has margin $\frac{3}{50} = 0.06$; the finite moment certificate has margin $\frac{3}{100} = 0.03$.

The dual witness is exact. Order the input facets by $F = [(0, 1); (0, -1); (2, 1); (2, -1); (-2, 1); (-2, -1)]$, $f = (\frac{4}{5}, \frac{4}{5}, 2, 2, 2, 2)$, and define $\eta_1 = (0, 0, 0, 0, \frac{1}{4}, \frac{1}{4})$, $\eta_2 = (0, \frac{1}{2}, 0, \frac{3}{10}, 0, 0)$, $\eta_3 = (\frac{1}{2}, 0, \frac{3}{10}, 0, 0, 0)$, satisfying $F^\top \eta_j = -n_j$ and $f^\top \eta_j = 1$. For a post-observation interval J_k put $a_k = \int_{J_k} (T - s) ds$ and $\mu_{j,k} = \lambda_j a_k \eta_j$. Before observation, stationarity holds because $\sum_j \lambda_j n_j = 0$; after observation, because $\lambda_j a_k n_j + F^\top \mu_{j,k} = 0$. Since $\sum_k a_k = \frac{1}{2}$, the Farkas contradiction is $\lambda^\top (\beta + e) + \mu^\top f_{\text{blk}} = -\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$ — an exact rational certificate. The primal witness is equally exact: hold the pre-observation blocks at 0 and the post-observation blocks of mode j at $-n_j$; the endpoint row then reads $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = \frac{3}{100}$, and every other row lies at or below the same value. Three structural facts make this transparent: the velocity and non-critical position rows are continuously safe at the witness ($\|v_\theta(t)\|_\infty \leq 1$ against the $\frac{6}{5}$ boundary; $n_i^\top n_j < 0$ for $i \neq j$ with the witness trajectory on the n_j ray), so their row values $\mathcal{F}_a(\hat{\pi}) - \bar{e}_a$ are nonpositive; the endpoint critical rows attain $\rho = \frac{3}{100}$ with equality; and the remaining critical rows interpolate monotonically between the two (the braking profile $\zeta(t) = \frac{39}{25} + (t - \tau) - \frac{(t - \tau)^2}{2}$ increases to its endpoint maximum) — these are verified row-by-row, with the campaign script checking all 465 rows exactly at this mesh and archiving the slacks (Section 14).

Instance arithmetic, collected. The exact rational chain of this subsection is: the pooling weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$ with $\sum_j \lambda_j = 1$ and $\sum_j \lambda_j n_j = 0$; the witness margins $\beta_j = \frac{16}{25} - t_*$ and the value $\Gamma(\tau) = \tau - \frac{7}{50}$ in (10), so $\tau_{\max} = \frac{7}{50}$ with equality viable; the moment error $e = Th/4$, giving at $h = \frac{1}{10}$ the relaxed right-hand side $\beta + e = -\frac{53}{100}$, the dual Farkas value $-\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$, and the primal endpoint row $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = +\frac{3}{100}$; and the law $\rho(h) = \frac{3}{50} - Th/4$ on every uniform aligned mesh (the level unrestricted; negative on coarser non-uniform aligned meshes). Every identity is an exact rational statement, machine-verified in Section 14.

Solver-assisted, exactly certified mesh study. For each mesh the program (5) is built at its claimed dimensions and solved by HiGHS; optimality is then certified in exact rational arithmetic by the witness pair above (primal

feasibility of every row; dual stationarity and dual objective $-\rho(h)$), so the duality gap closes exactly and the solver's returned value is a computational confirmation of a proved optimum, not an observation. Table 1 reports the campaign: exact values $\rho(h) = \frac{3}{50} - Th/4$ (nonnegative on this family), solver values, iteration counts, and wall-clock times as measured on the verification machine.

On the audited family of uniform meshes $h \in \{\frac{1}{5}, \frac{1}{10}, \frac{1}{20}, \frac{1}{50}, \frac{1}{100}\}$ (each dividing both τ and T), the values obey the exact law $\rho(h) = \frac{3}{50} - Th/4$ for $h < \frac{1}{5}$ and vanish at $h = \frac{1}{5}$: the whole refinement gap is the control-moment error $Th/4$ of the endpoint rows: the finite value lies below the continuous margin by exactly that error and converges to it from below as $h \downarrow 0$ ($\rho \leq J$ always — the program is a relaxation; Table 1 itself shows $0.03 < 0.06$). The $\max\{0, \cdot\}$ in the table's law is inactive on this family: for $h \leq \frac{1}{5}$ the raw value $3/50 - Th/4$ is already nonnegative, vanishing at $h = \frac{1}{5}$; the level is unrestricted, and coarser non-uniform meshes record negative values (below). For the record, the instance's hypothesis constants are $R_U = 1$ (the hexagon's circumradius), $V_U = \tau + 1 = \frac{6}{5}$ (the endpoint kernels are unit-slope ramps of height t^*), $\delta_\beta = 0$ (β is computed exactly; no stored-scenario error), and $L = T + 1 = \frac{11}{5}$ — the kernel difference $k_t - k_{t'}$ has L^1 norm at most $T|t - t'|$ (it is *not* supported on $[t', t]$: the position rows also differ by $(t - t')n_j$ on $[0, t']$; the integral bound is what (6) consumes), while $|\beta_t - \beta_{t'}| = |t - t'|$; the sandwich (8) thus bounds $J \leq \rho(h) + Th/4 + \frac{11}{5}h_t$, while the witness pair proves the realized gap is exactly $Th/4$. The $\max \bar{e}_a$ form of (8) is load-bearing: with the loose but verified choice $\bar{e} = e + \frac{1}{2}$ the program returns $\rho = -\frac{47}{100}$ at $h = \frac{1}{10}$, and the certified bound $J \leq \rho + \bar{e} + \delta_\beta + Lh_t = \frac{17}{100} \geq \frac{3}{50}$ stays valid, while the uniform special case $R_U V_U h_u$ would claim $J \leq -\frac{6}{25}$ — false at $J = \frac{3}{50}$. The domain is larger than the campaign: for every uniform aligned mesh (h dividing both τ and $T - \tau$) the same primal-dual witness pair proves $\rho(h) = \frac{3}{50} - Th/4$ — the dual objective $\sum_k a_k$ telescopes to $(T - \tau)^2/2$ independently of h — confirmed exactly at the sixth aligned mesh $h = \frac{1}{15}$, where $\rho = \frac{1}{25}$ ($Q = 48, 687$ inequalities). No floor is claimed off the uniform family: on an aligned mesh with unequal blocks the endpoint moment error is $\sum_J |J|^2/4$, and the allowed two-block mesh $[0, \frac{1}{5}], [\frac{1}{5}, \frac{6}{5}]$ gives $\rho = \frac{3}{50} - (\frac{1}{100} + \frac{1}{4}) = -\frac{1}{5}$, certified by the same witness pair (primal row $-\frac{1}{2} + \frac{14}{25} - \frac{26}{100}$; dual $\frac{1}{2} + (\beta + \frac{26}{100}) = \frac{1}{5} > 0$). For affine kernels the block moment error is exactly $|J|^2/4$ per block, so on this instance $e = \sum_J |J|^2/4$ on *every* mesh, uniform or not, and equal blocks minimize $\sum_J |J|^2$ at fixed count: time-refinement here has no conflict structure to target. The conflict-interval decomposition (12) is a different axis — it prices *information* cell splits, not control-block splits — and it is the axis the ladder subsection below reads prescriptively. All dimension counts ($mQ + 1$ variables; $Ms(N_t + 1) + p_U Q$ inequalities), all witness inequalities, and the closed-form law are machine-verified in exact rational arithmetic by the verification scripts of Section 14, which regenerate Table 1 verbatim and re-run the solver campaign.

h	blocks Q	variables $mQ+1$	rows $Ms(N_t+1) + p_U Q$	exact $\rho(h)$	solver ρ	iterations	time (s)
0.20	16	33	243	0	0.000000000	36	0.024
0.10	32	65	465	$\frac{3}{190} = 0.030$	0.030000000	78	0.014
0.05	64	129	909	$\frac{200}{200} = 0.045$	0.045000000	141	0.043
0.02	160	321	2241	$\frac{27}{590} = 0.054$	0.054000000	501	0.252
0.01	320	641	4461	$\frac{57}{1000} = 0.057$	0.057000000	1062	1.015

Table 1: Solver-assisted, exactly certified mesh study on the three-branch instance ($\tau = \frac{1}{5}$, $T = \frac{6}{5}$): program dimensions, exact optimal values, returned values (agreement to at most 4.2×10^{-17}), simplex iterations, and wall-clock time (single runs, scipy HiGHS dual simplex, on the verification machine; iteration counts and times may vary with solver version and settings; “rows” counts LP inequalities — safety plus actuator, with $s = 7$, $p_U = 6$, $N_t = T/h$). Optimality of each exact value is certified by the exact primal–dual witness pair, independent of the solver. At $h = 0.2$ the witness is an exact optimality witness of value zero; at the four finer meshes it is a positive obstruction certificate. The exact continuous value is 0.06; the three-label support of each positive certificate is an instance-specific fact of this symmetric campaign, distinct from the general $r + 1$ bound.

7. Computation, complexity, and independent verification

With Q information–time blocks and $N_t + 1$ time labels, the program has $mQ + 1$ variables and at most $Ms(N_t+1) + p_U Q$ inequalities (p_U input-polytope facets); it is polynomial-time in these dimensions and the encoded rational data size (weakly polynomial bit complexity) for the exact rational benchmark programs, and for interval-enclosed inputs once the enclosures are fixed — conditional on the validated coefficient bounds and certified row integrals. Initial-state dependence is affine, so no vertex enumeration of P_θ is required: at each label solve $\max_{\xi \in P_\theta} c_i^\top X_\theta(t, 0)\xi$ and store a maximizing or near-maximizing point — the verifier needs only membership of the stored point in P_θ , not global optimality; near-optimality bears on the convergence guarantee, not on a positive certificate’s validity. The same holds for stored disturbance signals.

For rational outer rows $\hat{A}v \leq \hat{b}$, $F_{\text{blk}}v \leq f_{\text{blk}}$, a residual-bounded certificate test suffices: with $\lambda, \mu \geq 0$, $r = \hat{A}^\top \lambda + F_{\text{blk}}^\top \mu$, $c = \lambda^\top \hat{b} + \mu^\top f_{\text{blk}}$, the condition

$$c + \sum_j h_U(-r_j) < 0 \quad (11)$$

is sufficient, since any feasible v satisfies $-\sum_j h_U(-r_j) \leq r^\top v \leq c$: the upper bound is the multiplier combination, the lower bound is termwise minimization over U , and the two displayed inequalities make the sign convention auditable. The actuator rows $F_{\text{blk}}v \leq f_{\text{blk}}$ already represent $v_j \in U$; their multipliers in the residual test are redundant but legitimate, and the verifier uses both paths as independent cross-checks. All integration, rounding, and model-enclosure errors must be included before applying the test, and uncertain row data must be relaxed toward a larger feasible controller set: overestimating the worst disturbance contribution can create a false obstruction — the sign convention of the $\beta_a^+ \geq \beta_a$ relaxation of Section 5: a nonmaximizing stored disturbance gives a larger β_a^+ , hence a weaker constraint, which is the sound direction for a lower bound on $J_{\mathcal{I}}(T)$. A deletion-based sparsification attempts supported labels for deletion one at a time in a fixed order, keeping a deletion only if the prescribed target margin survives the feasibility re-solve; it terminates with an irredundant (not claimed minimal) supported label set, at one feasibility solve per safety row; exact rank compression requires verified rank, and floating-point singular-value truncation must be charged

to the error budget. If $J_{\mathcal{I}}(T) = \gamma > 0$, a mesh with $\max_a \bar{e}_a + R_D V_D h_d + \delta_{\text{init}} + \delta_{\text{num}} + Lh_t < \gamma$ guarantees detection — an explicit margin-dependent resolution bound (mesh resolution as a function of the margin γ), not a certificate-size bound in general; the uniform choice $\bar{e}_a \equiv R_U V_U h_u$ gives the announced special case.

8. Information–time rank: input dimension does not bound the witnesses

Proposition 5 (arbitrarily large minimal obstructions with one input). *For every integer $q \geq 1$ there is a continuous-time system with scalar input $u(t) \in [0, 1]$ and $q + 1$ indistinguishable modes such that every proper sub-belief is viable, the full $(q+1)$ -mode belief is not, and every obstruction certificate must involve all $q + 1$ modes.*

Proof. Let $T = q$, $I_j = [j - 1, j]$. Mode $j \leq q$ carries the scalar state x obeying $\dot{x} = \mathbf{1}_{I_j}(t)(u - 1)$, $x(0) = 0$; mode $q + 1$ carries $\dot{x} = -u$, $x(0) = q - \varepsilon$, $0 < \varepsilon < 1$. Require $0 \leq x \leq q$ — the terminal bounds are the binding ones, each x being monotone along its mode — and there are no observations. Writing $w_j = \int_{I_j} u$, safety requires $-w_j \leq -1$ for $j \leq q$ and $\sum_j w_j \leq q - \varepsilon$; summing gives $0 \leq -\varepsilon$, a contradiction. If mode $q + 1$ is removed, $u \equiv 1$ is safe; if mode $j \leq q$ is removed, $u = 0$ on I_j and $u = 1$ elsewhere is safe. A certificate supported on fewer than $q + 1$ modes would also certify the corresponding proper subbelief, contradicting its viability, so every obstruction certificate involves all $q + 1$ modes. Equal weights give certificate margin $\varepsilon/(q + 1)$; the kernel rank is q , and $q + 1$ witnesses are necessary. $\square$

A statement such as “at most $m + 1$ compatible states suffice to witness nonviability” (Helly’s theorem, Helly, 1923) is valid for convex common-action sets in $\mathbb{R}^m$ but false for a common blind control function: the correct dimension counts independent temporal and informational control decisions. This distinction is consequential for review intervals, delayed sensing, and certificate-size claims; its discrete counterpart is a scoping caveat of the finite audits of Abaee (2026c).

9. Refinement erosion and oracle recourse

Suppose a coarse information cell C is split into finer cells $C_1, \dots, C_k$. Because $\phi_U(g_1 + \dots + g_k) \geq \sum_j \phi_U(g_j)$, the

same witness satisfies $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, with the exact difference

$$\int \left[\phi_U \left(\sum_j g_{C_j} \right) - \sum_j \phi_U(g_{C_j}) \right] dt. \quad (12)$$

For $U = [-\bar{u}, \bar{u}]^m$ this is $\bar{u} \int [\sum_j \|g_{C_j}(t)\|_1 - \|\sum_j g_{C_j}(t)\|_1] dt$: the certificate identifies the precise time intervals on which indistinguishable branches demand conflicting control effort. Equation (12) is the change in the same witness's margin under refinement, not a claim about the optimized viability value: a coarse cell can strengthen an obstruction, and splitting it can dissolve the certificate while the system's verdict is unchanged. The inequality is proved above by pointwise minimization; on rational instances it is additionally exactly checkable, and the verification record confirms $\phi_U(g_1 + g_2) \geq \phi_U(g_1) + \phi_U(g_2)$ on 3,240 deterministic rational pairs for the hexagonal U of Section 6 as a software regression test, not as evidence for the identity. The discrete counterpart of this axis is the master monotonicity of the worked-systems companion (Abaee, 2026c): its three axes (observations, policies, memory) enlarge kernels exactly as refinement here tightens the sandwich, and positivity is non-monotone on both sides.

The exact theorem assumes exogenous mode observations. A useful sound extension does not: if finitely many admissible initial-state/disturbance scenarios are provably indistinguishable until τ , give the controller after τ separate arbitrary control signals for every stored scenario — an oracle relaxation of the true policy class. Scenarios are stored initial-state/disturbance pairs, indistinguishable through τ in the sense that they generate identical observation histories; after τ the oracle reveals the realized scenario, and separate controls are granted even where scenarios remain observationally identical. The soundness direction: every real output-feedback policy induces one common pre- τ control, the oracle class contains every such policy, and nonviability in the enlarged class transfers down. If

$$\int_0^\tau \phi_U \left(\sum_\ell \lambda_\ell k_\ell \right) dt + \sum_\ell \int_\tau^T \phi_U(\lambda_\ell k_\ell) dt - \sum_\ell \lambda_\ell \beta_\ell > 0, \quad (13)$$

with β_ℓ computed from the explicitly stored scenario, then no actual output-feedback policy can be safe. This covers arbitrary sensing after a known blackout; it is sound but generally incomplete.

A third extension replaces exact scenario storage — and the exact revelation of the audited model — by a certified *belief cell*: a set $C \subseteq \Theta \times \mathcal{X}$, carried with a polytope enclosure, that the observation history certifies contains the true mode-state pair. — a set-valued observer with a guaranteed error radius, in the sense of the bounding approaches to estimation (Milanese et al., 1996).

Proposition 6 (belief-cell certificates). *Endow each row a with the cell label $\beta_a^C = \sup_{(\theta, x) \in C} \beta_a(\theta, x) + \delta_C$, the stored-scenario label replaced by its worst case over the cell, with $\delta_C \geq 0$ the certified enclosure radius of the position part (the position rows are unit). Then: (i) program (5) under the cell labels remains a linear program (conic wherever cell support functions enter), and the sandwich (8) and part (ii)'s soundness hold verbatim with the cell-filtration value in the middle: every policy adapted to the cell filtration*

is relaxed, and a positive value certifies nonviability of the original system. (ii) Splitting C into $C_1, \dots, C_k$ erodes a cell witness by exactly (12), charged cell by cell, and refinement to singletons recovers the stored-scenario labels (13): observation design is the same refinement axis as mesh design. (iii) Margins are radius-bounded on both sides: a primal witness whose worst row slack is s remains a viability witness for every $\delta_C \leq s$, and a dual certificate of margin Γ survives every $\delta_C < \Gamma$ (the labels' weights sum to one, so radius δ_C moves the margin to $\Gamma - \delta_C$).

Proof. (i) a cell-adapted policy must keep every certified pair in C safe, so the cell-sup rows relax exactly the cell-adapted class from above; the sandwich argument uses only the relaxation direction and the enclosure bound, and the upper side's charges are realized by held controls as before. (ii) the cell label is a pointwise supremum, so splitting replaces each sup by the maximum of sub-cell sups; the resulting gap is the displayed subadditivity $\phi_U(\sum_j g_{C_j}) - \sum_j \phi_U(g_{C_j})$ of (12), and singleton cells restore the stored labels of (13). (iii) shifting the true state inside a radius- δ_C ball translates every affine row value by at most δ_C : a witness with slack s absorbs it, and the dual combination's value drops by $\delta_C \sum_j \lambda_j = \delta_C$. $\square$

On the instance the belief cells are the compatible-mode sets, and the audited delay hierarchy is exactly the cell-refinement family: the full cell $\{1, 2, 3\}$ obstructs past $\frac{7}{50}$, the two-mode cells $\{1, 2\}$ and $\{2, 3\}$ are viable to $\frac{15-\sqrt{183}}{6}$ and $\frac{10-\sqrt{58}}{6}$, and the singleton cells never obstruct (Proposition 7). Noisy state cells price in the same units: at $\tau = \frac{1}{5}$ the pair viability witnesses tolerate cell radius up to their worst row slacks $\frac{31}{1250}$ and $\frac{81}{1250}$, and the triple's obstruction certificate survives any state-cell radius $\delta_C < \Gamma(\frac{1}{5}) = \frac{3}{50}$, the margin moving to $\tau - \frac{7}{50} - \delta_C$. On the exact-computation companion's deadline instance the same law is exact: the radius- δ_C cell inherits the viability verdict precisely for $\delta_C \leq z_0 - (1 + T/2)$, with the boundary cells zero-slack (joint check archived; Abaee, 2026f).

10. Reading the certificate prescriptively: the shared-authority ladder

A positive certificate is not only a verdict; its weights, kernels, and facet multipliers encode what a controller would have to change. On the instance the decoding is exact end to end.

Proposition 7 (shared-authority ladder). (i) *The pooling weights satisfy $\sum_j \lambda_j n_j = 0$ with $\lambda_j > 0$; hence for every common control $w \in U$ some branch has $n_j^\top w \geq 0$: no common control brakes all three branches, and the pooled blind-window kernel — the certificate's zero charge on $[0, \tau]$ — is exactly this unbrakeability. (ii) The optimal shared braking authorities $t^*(B) = \max_{w \in U} \min_{j \in B} (-n_j^\top w)$ are $t^*(\{1, 2\}) = t^*(\{1, 3\}) = \frac{2}{5}$, attained at $u = (-\frac{2}{5}, -\frac{4}{5})$ and its v_2 -reflection, and $t^*(\{2, 3\}) = \frac{3}{5}$, attained at $u = n_1$: the blind-window controls of the pair policies of Section 6 are exactly these optima. (iii) Braking at authority t^* through the blind window and $u = -n_j$ afterward holds the critical-position peak at $\frac{93}{50} + (1 - t^*)\tau + \frac{(t^*)^2 - t^*}{2}\tau^2$, increasing in τ on $[0, 1/t^*]$; the pair delay thresholds — where the*

peak reaches 2 — are $\tau_{12} = \tau_{13} = \frac{15-\sqrt{183}}{6} \approx 0.245$ and $\tau_{23} = \frac{10-\sqrt{58}}{6} \approx 0.397$, against the triple's $\tau_{\max} = \frac{7}{50}$, with singletons unbounded (the full-information peak $\frac{93}{50}$ is delay-independent). For every τ at or below its threshold the critical position stays ≤ 2 with the velocity rows strict; the verification script verifies the full seven-facet scan exactly at rational surrogates below each threshold ($\tau = \frac{6}{25}$ and $\tau = \frac{39}{100}$).

Proof. (i) is the pooling identity itself: $\sum_j \lambda_j (n_j^\top w) = (\sum_j \lambda_j n_j)^\top w = 0$ cannot have every summand strictly negative. (ii) The objective $\max(n_i^\top w, n_j^\top w)$ is piecewise linear, so its minimum over the hexagon is attained at a vertex or on the equal-projection segment $(n_i - n_j)^\top w = 0$. On that segment both authorities coincide: $w = -c(1, 2)$ for $\{1, 2\}$ (the facet $|u_2| \leq \frac{4}{5}$ binds at $c = \frac{2}{5}$) and its mirror for $\{1, 3\}$; $w = (c, 0)$ for $\{2, 3\}$ (the vertex n_1 binds at $c = 1$); the vertices themselves give at most $\frac{7}{25}$ for the mixed pairs and $\frac{3}{5}$ for $\{2, 3\}$. (iii) Along the optimal schedule the window exit state is $\frac{34}{25} + \tau - t^* \tau^2/2$ at velocity $1 - t^* \tau$ on the critical ray, and full braking adds $(1 - t^* \tau)^2/2$; collecting terms gives the displayed quadratic, whose τ -vertex sits at $1/t^* \geq 1$, so it increases on the audited range, and the thresholds are its exact roots in $\mathbb{Q}(\sqrt{183})$ and $\mathbb{Q}(\sqrt{58})$ ($6\tau^2 - 30\tau + 7 = 0$ and $6\tau^2 - 20\tau + 7 = 0$). The facet scans are exact rational evaluations at the stated candidates. $\square$

The prescriptive reading follows directly. The obstruction is removed by shortening the observation window below the relevant rung (the pairs already tolerate $\tau = \frac{1}{5}$ under shared braking alone — the audited delay is strictly inside both pair rungs), by adding observers (each information refinement deletes rungs, and the conflict-interval decomposition (12) prices exactly which splits pay), or by reallocating authority where the certificate charges it (the post-window controls already brake at the hexagon's face $-n_j \in U$, so the binding resource is the shared window, not the post-observation actuator). The dual witness agrees: the η_j place their mass on the braking facets $\mp n_j$, and the primal witness brakes every mode to rest — the certificate is the blind window's unbrakeability, propagated to the deadline. The decode is quantitative: each redesign lever carries an exact rate.

Proposition 8 (redesign sensitivities). *On the instance, dilate the post-revelation control set to aU (the audited hexagon is $a = 1$) and let the observation time τ vary. (i) Timing: $\Gamma_a(\tau) = \tau - \frac{7}{50} - \frac{a-1}{2}$: advancing observation by δ buys margin δ , at every a . (ii) Post-revelation authority: the exact delay threshold is $\tau_{\max}(a) = \frac{7}{50} + \frac{a-1}{2}$ (equality viable: the safety set is closed), so each unit of dilated braking authority buys exactly half a unit of delay tolerance. (iii) Shared-window authority: the pooled pre-observation kernel is $\sum_j \lambda_j k_j \equiv 0$ as a vector identity and the witness's pre-observation stationarity is the pooling identity alone, so the margin is invariant under any enlargement of the blind-window control set, however large: no shared pre-observation authority moves the threshold. (iv) Velocity budget: the printed dual places no multipliers on the velocity rows (their primal slacks are strictly positive at the witness), so any relaxation of $|v_1|, |v_2| \leq \frac{6}{5}$, up to deleting the rows entirely, leaves the witness feasible with value*

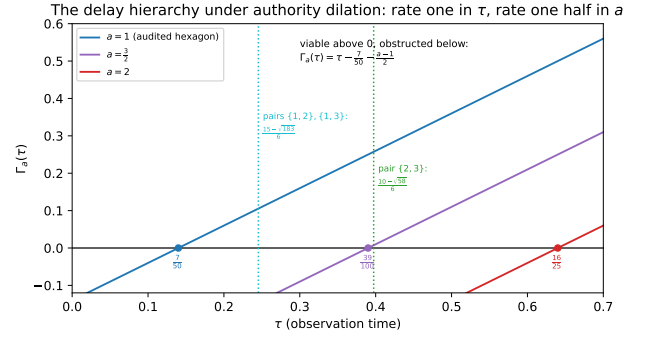


Figure 2: The delay hierarchy under authority dilation: $\Gamma_a(\tau) = \tau - \frac{7}{50} - \frac{a-1}{2}$ crosses zero at $\frac{7}{50}$, $\frac{39}{100}$, $\frac{16}{25}$ for $a = 1, \frac{3}{2}, 2$ (rate one in τ , rate one half in a); the pair delay thresholds $\frac{15-\sqrt{183}}{6}$ and $\frac{10-\sqrt{58}}{6}$ mark the shared-braking rungs of Proposition 7. Same assertion script.

$\tau - \frac{7}{50}$: the obstruction is positional, carried by braking liability against position facets.

Proof. Under $U_a = aU$ the support values scale, $\max_{u \in aU} \langle u, n_j \rangle = -a$, so the post-observation charge in (10) is $-\frac{a}{2}$ and $\Gamma_a(\tau) = -\frac{a}{2} - (\frac{16}{25} - \tau - 1) = \tau - \frac{7}{50} - \frac{a-1}{2}$; dual feasibility is preserved on the dilated facets $F_a = aF$, $f_a = af$ by $\eta_j/a \geq 0$, which satisfy $F_a^\top(\eta_j/a) = -n_j$ and $f_a^\top(\eta_j/a) = 1$; the threshold solves $\Gamma_a(\tau) = 0$, with equality viable as at $a = 1$. (iii) $\sum_j \lambda_j k_j = (t_* - s) \sum_j \lambda_j n_j = 0$ independent of the control set, and the printed dual uses no pre-observation input facets — stationarity there is $\sum_j \lambda_j n_j = 0$ alone. (iv) deleting rows removes no dual term: the witness's velocity-row multipliers are zero and its value is unchanged. $\square$

The rates order the redesign options exactly: observation timing at one, post-revelation authority at one half, shared-window authority and the velocity budget at zero.

11. The reference library

The finite side of the pipeline is packaged as **viacert**, a standard-library-only Python package (no third-party dependencies) using exact integer and rational arithmetic. Its design follows three principles. *Exactness*: the finite audits and exact certificate verifiers use integer and rational arithmetic; floating-point solver output is never used as a proof object. *Fidelity to the verified recursion*: the finite-horizon selector is the calculus's backward recursion (Abaee, 2026b) with the no-repeat protocol threaded as a parameter, matching the audited implementations rather than a reformulation. *Pluggability*: new systems enter through a **FiniteSystem** interface (states, actions, safe set, transition, null action); the audited systems — the shared two-coordinate audit system, the continuous benchmark caps, the certainty-equivalence law pair, and the review grid — ship as default instances.

The public surface comprises the recursion module (viability verdicts, kernels, and step-one witness extraction), the monitoring module (safe and common safe-action sets, the partition-form adequacy converse, maximal common-action sets, the partition census, the review-timing identity with its threshold convention, and the certainty-equivalence drift audit), and a command-line self-test of twelve audited identities — the four-cell kernel counts

24/26/25/28 with the meet identity, the census 15 total and 7 adequate partitions with maximal sets overlapping in (2, 2), the benchmark $Y^* = 27/5$, the drift bounds 21/100 and 0, the 48-cell delay identity with its boundary cell, and witness extraction — each a pointer to the published audits (Abaee, 2026b, 2026c). The inherited counts are software self-tests pointing to the published audits; they are not used in this paper’s main theorem or example, and their full context lives in the package documentation. In the certification workflow the library plays the enumerate-and-check role: finite instances are enumerated exactly, candidate designs are tested against the fibre criteria, and every claim is archived with the script that regenerates it. The library is not a reimplement of this paper’s bridge theorems: it carries the exact finite machinery they reduce from, while the bridge’s programs live in the campaign scripts of Section 14; consolidating those scripts as a `viacert.continuous` module is the library’s recorded next step, together with three recorded tooling directions: warm-started incremental refinement (re-solve on block splits seeded with the coarse dual, benchmarked against cold re-solve), an arbitrary-precision dual re-solve fallback for interval-coefficient programs (solve at working precision, detect dual-gap tightness, re-solve the tight dual in extended precision), and conflict-targeted block refinement where kernels are non-affine. Consolidation supersedes implementation paths only in future releases: the archived scripts and their audited identities remain the exact versions used for this article’s verification record. The title’s “reference library” is read as this finite-side package plus the certified campaign pipeline.

12. Scope delimitations

The exact theorem does not cover: control-dependent observations or active sensing; arbitrary continuous-state measurements; nonlinear dynamics without additional validated enclosures; nonconvex actuator sets without a relaxation gap (convexity is structural: block averages of admissible controls lie in $\text{conv } U$, and the weak-* compactness of Proposition 2 uses that U is a compact convex polytope); rate-constrained actuators without modifying the approximation; or unrestricted infinite-horizon completeness. For general sensing after a blackout, the oracle-recourse certificate is sound but incomplete — exactness in the audited instance belongs to the complete revelation at τ , not to the relaxation. For nonlinear systems, a uniform validated flow-error bound can be added to the row budgets, yielding a sound extension, not automatic completeness. Against the alternatives: full-information viability is necessary but not sufficient for the information-constrained question; initial-admissibility and tangency tests are silent on the instance’s obstruction, which is temporal-informational rather than pointwise-geometric; safety-before-first-observation tests are silent when a common safe blind control exists; exact finite safety games exist as discrete models, and what is needed to use them as sound abstractions of the present continuous systems is a justified continuous-to-finite bridge, which is what Theorem 4 provides; and naïve held-control infeasibility excludes only the held-control class. No general-purpose computational-calculus claim is made: the theorem is a certificate generator for the stated class, with complete-

ness only under refinement and semidecidability only with effectively computable bounds.

13. Conclusion

A sound and independently checkable certification pipeline under incomplete observation rests on three requirements: the finite controller class must be an outer relaxation while stored adversarial tests correspond to genuinely admissible scenarios; optimality must be certifiable independently of the solver; and certificate size must be understood in the information–time product rather than the input dimension. The bridge theorem delivers all three, the three-branch instance shows the resulting certificates detecting failure that full-information viability, blind-window exit tests, and held-control infeasibility all miss, and the campaign shows the solver returning the independently proved optima to numerical precision while exact witnesses close every gap. The reference library carries the exact finite machinery to new systems. What remains open is completeness beyond refinement: an inner approximation of the policy class that is computationally effective for general sensing is not known, and the oracle-recourse extension is the current sound surrogate.

Three further extension directions are recorded, each a construction of its own rather than an increment of the present framework; all lie outside the audited class delimited in Section 12. The cell-conditioned lift of labels, kernels, and costs for general measurement channels — the passage from history partitions to general measurement maps — is carried in Proposition 6, with the instance’s compatible-mode hierarchy and radius bounds as the audited instantiation. (i) Stochastic disturbances: replacing support-function rows with second-moment bounds turns the finite program into a conic one (second-order cone or semidefinite), with exact rational witnesses replaced by conic dual certificates; the exact rational optima of the present campaign remain the benchmark such relaxations must meet. (ii) Decentralized and networked controller sets, infinite-horizon and periodic regimes, and nonconvex or piecewise-smooth actuators: each alters a structural input of the bridge theorem (the weak-* compactness of label strategies, finite-horizon completeness, or the convexity of the actuator set), and is therefore a new construction rather than a parameter change. (iii) A cross-tool comparison study that treats the audited instance as a common benchmark — exact finite safety games, interval-reachability toolchains, SMT-backed barrier synthesis — with the certificate file as the exchange format; the tooling directions recorded in Section 11 and the independent certificate-file parser of Section 14 are its computational prerequisites.

14. Verification methods

Every reported mathematical benchmark value and every table entry is regenerated by deterministic scripts, exact rational arithmetic unless stated as solver output, standard library plus the solver interface only. Three scripts and one package cover the claims, with one master entry point chaining them. The exact verification record rederives the instance’s certificates — delay threshold, margins, witnesses, erosion identity, rank construction — in 51

exact checks. The campaign script rebuilds the program (5) at each of the five meshes, runs HiGHS, and certifies optimality by the exact primal–dual witnesses, closing the duality gap; it re-derives the mesh law and all dimension identities (nine checks). The library self-test runs the twelve audited identities. A master script chains the three, then recomputes independently: the mesh values from the closed law $3/50 - Th/4$ (uniform aligned meshes), the Farkas contradiction $-\frac{53}{100} + \frac{1}{2} = -\frac{3}{100}$, the primal row value $\frac{3}{100}$, the margins $\frac{3}{50}$ and $\frac{3}{100}$, the moment error $e = \frac{3}{100}$, the relaxed right-hand side $-\frac{53}{100}$, the delay threshold $\tau^* = \frac{7}{50}$, the blind-window safety value $\frac{39}{25}$, and every headline number in the text; independence here means a separate formula path from the subordinate scripts, not a separate implementation — the independent certificate-file parser is shipped (`certificate_exchange_v1.py`) and validates the archived witnesses of this paper and of the dual-certificate companion (Abaee, 2026g) exactly, in either direction, alongside the tooling directions of Section 11. The two figures are generated by `paper2_comp_certification_figures_v1.py` (recompute-then-assert: every plotted peak, threshold, and crossing is recomputed in exact rational arithmetic and asserted against the printed values before the figure is written; ten assertions). Timings are single runs on the verification machine (Linux x86_64, Intel Xeon @ 2.60 GHz, Python 3.13, SciPy 1.17 HiGHS dual-simplex interface, solver defaults otherwise).

References

- Abaee, A., 2026b. An obstruction calculus for viability under incomplete observation. Manuscript submitted for publication.
- Abaee, A., 2026c. Worked systems for the obstruction calculus: monotonicity, stationary policies, and decentralized factoring. Manuscript submitted for publication.
- Abaee, A., 2026g. The measure dual of the common-action obstruction, and the dynamic envelope. Manuscript submitted for publication.
- Chvátal, V., 1983. Linear Programming. A Series of Books in the Mathematical Sciences. W. H. Freeman, New York.
- Helly, E., 1923. Über Mengen konvexer Körper mit gemeinschaftlichen Punkten. J. Deutsche Math.-Ver. 32, 175–176.
- Milanese, M., Norton, J., Piet-Lahanier, H., Walter, E. (Eds.), 1996. Bounding Approaches to System Identification. Plenum Press, New York.
- Abaee, A., 2026f. Exact belief-state computation at scale: the four-parameter cube, Hamming adjacency, and the antichain census. Manuscript submitted for publication.
- Abaee, A., 2026a. Robust viability of the 2J3KL limit reference point under a surplus-production map: policy scoring, expansion, and when catch cannot help. Zenodo. <https://doi.org/10.5281/zenodo.22552060>
- Anderson, E.J., Nash, P., 1987. Linear Programming in Infinite-Dimensional Spaces: Theory and Applications. Wiley, Chichester.
- Aubin, J.-P., 1991. Viability Theory. Birkhäuser, Boston.
- Cardaliaguet, P., Quincampoix, M., Saint-Pierre, P.,

1999. Set-valued numerical methods for optimal control and differential games, in: Stochastic and Differential Games: Theory and Numerical Methods, pp. 177–247. Birkhäuser, Boston.

Hettich, R., Kortanek, K.O., 1993. Semi-infinite programming: theory, methods, and applications. SIAM Rev. 35(3), 380–429.

Sion, M., 1958. On general minimax theorems. Pacific J. Math. 8(1), 171–176.

Huangfu, Q., Hall, J.A.J., 2018. Parallelizing the dual revised simplex method. Math. Program. Comput. 10(1), 119–142.

Veliov, V.M., 1993. Sufficient conditions for viability under imperfect measurement. Set-Valued Anal. 1, 305–317.

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification scripts `paper2_computational_certification_v18_verification.py`, `paper2_nonstandard_verification.py`, `paper2c_lp_campaign.py`, and the `viacert` package are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`, at the commit pinned to this article’s submission); they reproduce all exact values, witnesses, bounds, and the solver campaign verbatim.

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